

Title

Expanded tuberculosis diagnostic algorithms for hospitalized HIV-positive patients: a pragmatic randomized trial protocol

Abstract

Rationale: Tuberculosis remains the leading cause of mortality in hospitalized persons with advanced HIV. Standard symptom-triggered diagnostic approaches may miss TB in atypical presentations. This protocol describes a pragmatic trial evaluating expanded diagnostic strategies. **Methods:** Hospitalized adults with HIV (≥ 18 years, no prior TB diagnosis or recent treatment) will be enrolled within 24 hours of admission across 11 hospitals in Tanzania and Mozambique (target $n=1200$). Participants will be randomly assigned 1:1 to expanded diagnostic arm or standard-care arm. Expanded arm receives TB testing from sputum, stool, and urine specimens plus urine antigen testing, irrespective of symptoms. Standard arm receives symptom-based WHO-recommended sputum and urine antigen testing. All participants receive medical management per standard protocols. **Primary outcome:** proportion with microbiologically confirmed TB initiating treatment within 72 hours. **Secondary outcomes:** 8-week mortality, time from hospital admission to TB diagnosis, and diagnostic yield by specimen type. **Follow-up:** all participants followed through 8 weeks post-enrollment for mortality and TB diagnosis outcomes. **Blinding:** laboratory personnel blinded to diagnostic arm. **Study design:** pragmatic individually randomized controlled trial.